GAME THEORY & NASH EQUILIBRIA — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Game Theory & Nash Equilibria

Question 1

10.0 points

Examine the extensive form game tree in the diagram below for a sequential two-player game where Player 1 can choose between actions A and B, and Player 2 can then choose between X and Y after observing Player 1's action. Determine the subgame perfect Nash equilibrium by performing backward induction on this game tree.

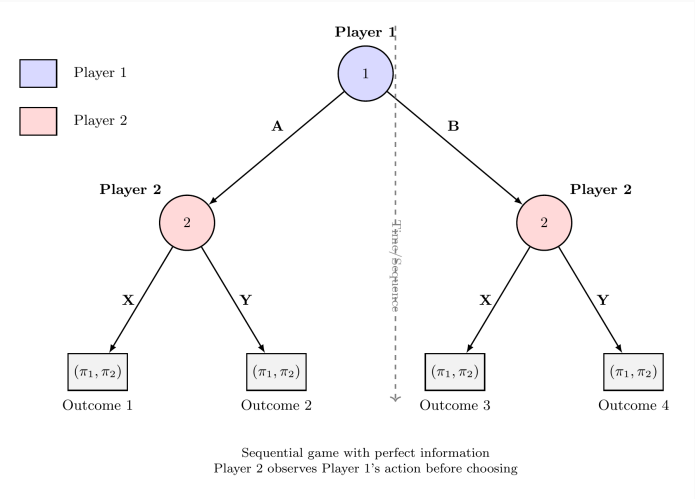

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Consider a payoff matrix for a Prisoner's Dilemma: | | Cooperate | Defect | |------- |----------|--------| | C | (3,3) | (0,5) | | D | (5,0) | (1,1) | Identify all Nash equilibria in this game and explain why they are Nash equilibria.

Question 3 10.0 points

Evaluate a mixed strategy Nash equilibrium in a coordination game given by the payoff table in the diagram below, where both players choose high or low effort: | | High | Low | |----------|-------|-------| | High | (2,2) | (0,3) | | Low | (3,0) | (1,1) | Use the mixed strategy method to find the equilibria, and determine the probabilities with which each player should randomize their strategies.

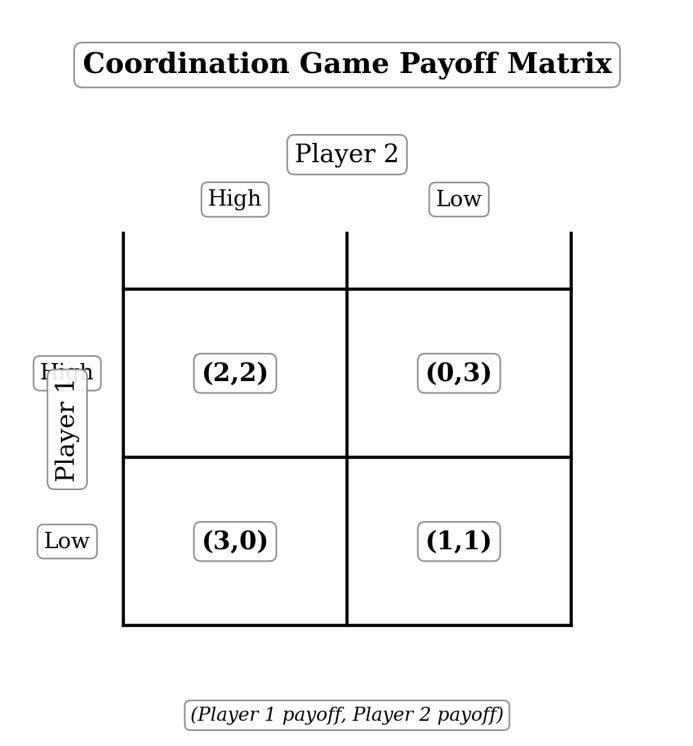

Fig. 3. Diagram for Question 3

Question 4 10.0 points

Perform backward induction on the game tree shown in the diagram below for a three-stage sequential game with perfect information. Each player has one possible optimal move at each decision node. Determine the optimal strategy for each player, and explain how backward induction aids in determining this strategy.

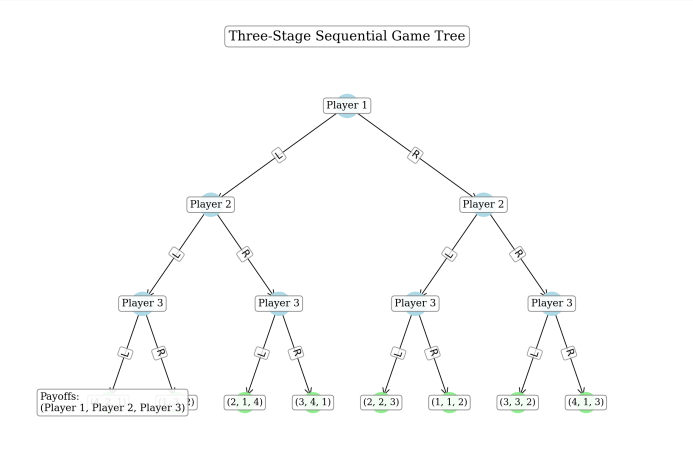

Fig. 4. Diagram for Question 4

Question 5 10.0 points

Given the Pareto frontier in the diagram below illustrating efficient vs inefficient outcomes, identify the Pareto efficient points and describe the trade-offs at inefficient points. Explain how any shift in the curve can alter efficiency in allocation.

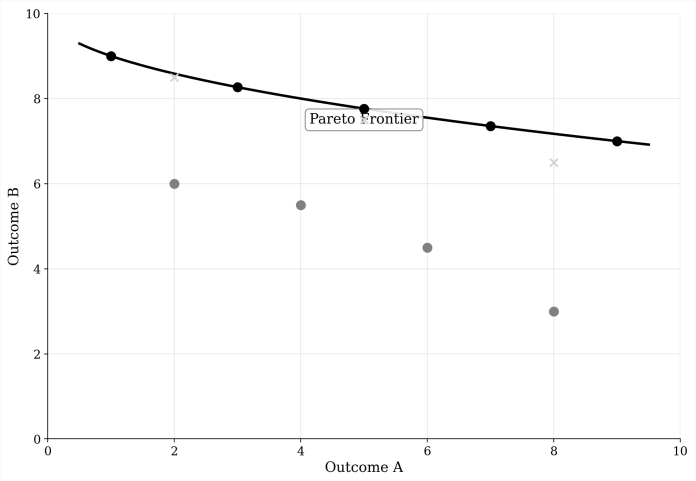

Fig. 5. Diagram for Question 5